

UNIT 03

STATISTICAL INFERENCE, EXPLORATORY DATA ANALYSIS AND THE DATA SCIENCE PROCESS

BEGINNER-FRIENDLY STUDENT NOTES

Syllabus coverage:

Populations and Samples • Statistical Models • Probability Distributions • Statistical Inference • Parametric and Non-Parametric Methods • Distance Metrics • EDA Tools and Philosophy • Data Science Process

Ask a clear question → Explore the evidence → Measure uncertainty → Communicate honestly

A. How to Use These Notes

- **Begin with the vocabulary:** understand population, sample, variable, parameter, statistic and probability before studying tests.
- **Do the small calculations:** manual examples show what formulas and Python functions actually calculate.
- **Read graphs as questions:** do not only name a plot; state what pattern, difference or unusual value it reveals.
- **Check assumptions:** a statistical method is useful only when its conditions are reasonable for the data.
- **Revise actively:** cover the answers and attempt the quick checks, MCQs and exam questions yourself.

STUDY RULE

For every concept, remember: meaning, purpose, assumptions, small example and interpretation.

B. Learning Outcomes

- Differentiate population, sample, census, parameter and statistic.
- Explain sampling error, sampling bias and why statistical inference is required.
- Recognize major types of statistical modelling and probability distributions.
- Explain estimation, confidence intervals, hypothesis tests and p-values at an introductory level.
- Select parametric or non-parametric methods according to data and assumptions.
- Calculate and interpret common distance and similarity measures.
- Use summary statistics, plots and graphs for exploratory data analysis.
- Explain the philosophy of EDA and follow the complete iterative data science process.

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1. Essential Statistics Basics

1.1 What is Statistics?

Statistics is the science of collecting, organizing, summarizing, analysing and interpreting data so that decisions can be made under uncertainty. Statistics does not remove uncertainty; it helps us describe and manage it.

Part	Main purpose	Examples
Descriptive statistics	Describe the data that were observed	Tables, graphs, mean, median, standard deviation
Inferential statistics	Use sample evidence to learn about a population	Estimation, confidence intervals, hypothesis tests

MEMORY AID

Descriptive = describe the observed data. Inferential = infer beyond the observed sample.

1.2 Data, Variables and Observations

Term	Meaning	Example
Observation / case	One measured unit or row	One student record
Variable	A characteristic measured for every case	Age, marks, department
Quantitative variable	Numeric values for which arithmetic is meaningful	Height, income, attendance
Categorical variable	Names or groups	City, blood group, course
Discrete variable	Countable values	Number of absences
Continuous variable	Values across an interval	Weight or time

1.3 Levels of Measurement

Scale	Meaning	Example	Suitable summaries
Nominal	Categories without order	Department, colour	Count, percentage, mode
Ordinal	Ordered categories; gaps not necessarily equal	Low, medium, high	Median, rank, percentages
Interval	Equal differences; no true zero	Temperature in Celsius	Mean, SD, differences
Ratio	Equal differences and a meaningful zero	Age, distance, income	All arithmetic summaries

KEY POINT

The type and measurement scale of a variable influence which graph, distance metric and statistical method should be used.

2. Populations and Samples

2.1 Basic Terms

Term	Simple meaning	Example
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Term	Simple meaning	Example
Population	The complete group we want to understand	All students in a university
Target population	The exact group to which conclusions should apply	All first-year students in 2026
Sample	A smaller observed subset of the population	300 selected students
Census	Collecting data from every population member	University-wide enrollment census
Parameter	An unknown numerical fact about the population	Population mean attendance
Statistic	A number calculated from the sample	Sample mean attendance

FROM POPULATION TO SAMPLE AND BACK TO INFERENCE

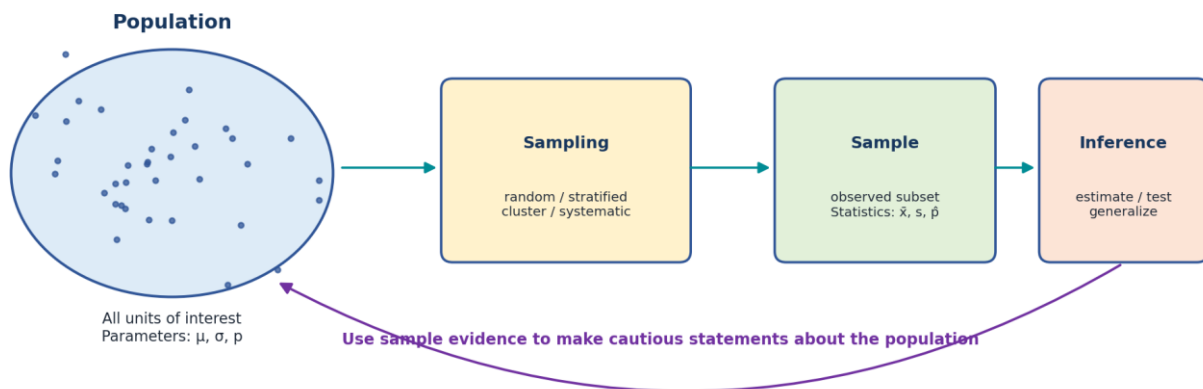


Figure 1: A sample supplies evidence used to learn about a wider population.

MEMORY AID

Population gives parameters. Sample gives statistics. Statistics are used to estimate parameters.

2.2 Why Use a Sample?

- A census may require too much time, money or effort.
- The population may be extremely large or continuously changing.
- Some measurements are destructive, such as testing the lifetime of a bulb until it fails.
- A carefully selected sample can provide useful estimates with measurable uncertainty.

2.3 Common Sampling Methods

Method	How it works	Simple example
Simple random	Every unit has a known equal chance	Randomly choose roll numbers
Systematic	Choose every k-th unit after a random start	Every 10th name on a list
Stratified	Divide into groups and sample from each	Sample separately from each department
Cluster	Randomly select whole groups	Select colleges, then survey students inside them
Convenience	Use the easiest available cases	Survey students present in one class
Volunteer	People choose to respond	Open online opinion poll

CAUTION

Convenience and volunteer samples are easy to collect but may not represent the target population. A large biased sample can still produce a wrong conclusion.

2.4 Sampling Error and Bias

Concept	Meaning	Can a larger sample solve it?
Sampling error	Natural difference caused by observing only a subset	Usually reduced, if sampling is sound
Selection bias	Some population members are systematically over- or under-represented	Not automatically
Measurement bias	The measuring tool or question systematically distorts values	No; improve measurement
Non-response bias	Respondents differ from non-respondents	Not automatically

EXAMPLE

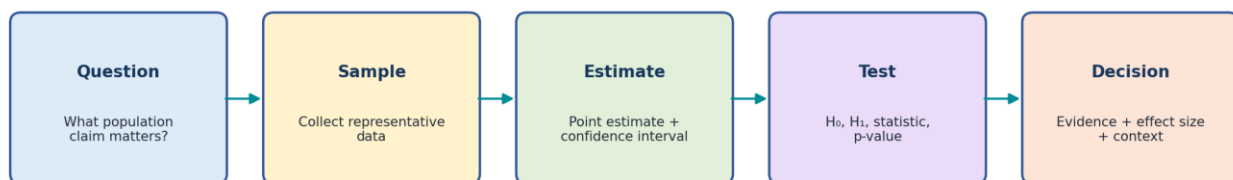
Tasting one spoon from a well-stirred pot shows sampling variation. Tasting only the top oily layer shows selection bias.

3. Statistical Inference Basics

3.1 What is Statistical Inference?

Statistical inference is the process of using a sample to estimate, compare or test statements about a population. Because another sample would give a slightly different result, inference must include uncertainty.

STATISTICAL INFERENCE WORKFLOW



Inference quantifies uncertainty; it does not eliminate uncertainty.

Figure 2: From a population question to evidence, uncertainty and interpretation.

3.2 Point Estimate and Standard Error

A point estimate is one sample value used to estimate a population parameter. The sample mean estimates the population mean. The standard error measures how much the estimate would vary across repeated samples.

FORMULA

Estimated standard error of a sample mean: $SE = s / \sqrt{n}$, where s is sample standard deviation and n is sample size.

Example: delivery times are 28, 32, 31, 35 and 29 minutes. The sample mean is 31 minutes. If the sample standard deviation is about 2.74, then $SE = 2.74 / \sqrt{5} =$ about 1.23 minutes.

Measure	What it describes
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Measure	What it describes
Standard deviation	Variation among individual observations
Standard error	Uncertainty or sample-to-sample variation of an estimate

3.3 Confidence Interval

A confidence interval gives a plausible range for a population parameter by combining an estimate with a margin of error. A narrower interval means greater precision, provided the method assumptions are reasonable.

GENERAL FORM	Estimate $\pm$ critical value $\times$ standard error
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Example: sample mean = 72, standard error = 2 and approximate 95% critical value = 1.96. The interval is $72 \pm 1.96 \times 2 = [68.08, 75.92]$.

INTERPRETATION	A 95% confidence procedure is designed so that about 95% of intervals produced in repeated valid samples contain the true parameter.
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3.4 Hypothesis Testing

A hypothesis test evaluates whether the observed data are reasonably compatible with a reference claim called the null hypothesis.

Term	Meaning
Null hypothesis (H_0)	Reference or no-effect claim assessed by the test
Alternative hypothesis (H_1)	Competing claim supported when evidence against H_0 is strong
Test statistic	A standardized measure of disagreement between data and H_0
p-value	Under H_0 and the test assumptions, probability of a result at least as extreme as observed
Significance level (α)	A chosen threshold such as 0.05 for deciding whether evidence is statistically significant
Effect size	The magnitude of a difference or relationship

CAUTION	A p-value is not the probability that H_0 is true. A small p-value also does not prove that an effect is practically important.
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3.5 Type I and Type II Errors

Actual condition / decision	Do not reject H_0	Reject H_0
H_0 is true	Correct decision	Type I error: false positive
H_0 is false	Type II error: false negative	Correct detection; called statistical power

MEMORY AID	Type I = false alarm. Type II = missed detection.
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4. Types of Statistical Modelling

4.1 What is a Statistical Model?

A statistical model is a simplified mathematical description of how data may be generated. It represents systematic patterns plus unexplained random variation. A model is selected to answer a specific question; it is not a perfect copy of reality.

EXAMPLE

In $y = b_0 + b_1x + \text{error}$, $b_0 + b_1x$ represents the expected relationship and error represents unexplained variation.

4.2 Models Classified by Purpose

Purpose	Main question	Typical result	Example
Descriptive	What happened or what pattern exists?	Summary, graph, segment	Describe marks by department
Inferential / explanatory	What population relationship or effect is supported?	Estimate, confidence interval, test	Estimate mean difference between teaching methods
Predictive	What will happen for a new case?	Predicted value/class and performance metric	Predict student pass/fail
Prescriptive	What action should be taken?	Recommended decision under constraints	Choose an intervention plan

4.3 Common Model Families

Model family	Suitable outcome / structure	Examples
Regression	Continuous numeric target	Linear regression, polynomial regression
Classification	Categorical target	Logistic regression, decision tree, KNN
Count model	Non-negative event counts	Poisson regression
Time-series model	Time-ordered observations	Moving average, ARIMA
Survival model	Time until an event	Kaplan-Meier, Cox regression
Unsupervised model	No labelled target	Clustering, PCA

4.4 Model Assumptions and Evaluation

- **Assumptions:** conditions about independence, distribution, linearity or equal variance that support valid interpretation.
- **Fit:** how well the model represents observed data.
- **Generalization:** how well it works on new data, not merely the training data.
- **Interpretability:** whether humans can understand the relationship or decision.
- **Simplicity:** a more complex model should be used only when it gives meaningful improvement.

KEY POINT

The best model is not always the most complicated model. It must match the question, data, assumptions and intended use.

5. Types of Probability Distributions

5.1 Probability Basics

Probability measures uncertainty using values from 0 to 1. A random variable assigns a number to an uncertain outcome. A probability distribution lists or describes the possible values and their probabilities.

Concept	Discrete random variable	Continuous random variable
Possible values	Countable values	Any value within intervals
Examples	Number of defects, pass/fail	Height, temperature, waiting time
Function	Probability mass function (PMF)	Probability density function (PDF)
Exact-value probability	May be greater than 0	Normally 0; use intervals

COMMON DISCRETE AND CONTINUOUS PROBABILITY DISTRIBUTIONS

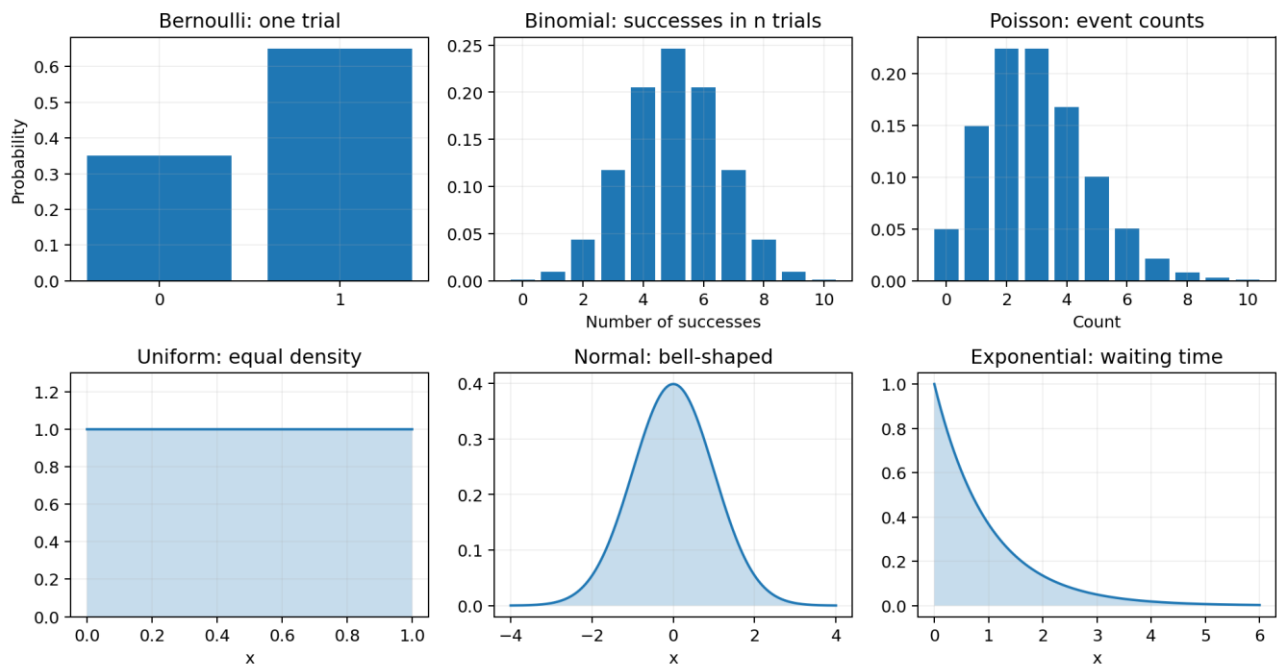


Figure 3: Common distribution shapes and the situations they can represent.

5.2 Mean, Variance and Shape

- **Expected value / mean:** the long-run centre or probability-weighted average.
- **Variance:** average squared distance from the mean.
- **Standard deviation:** square root of variance, measured in the original unit.
- **Skewness:** asymmetry of a distribution. Right-skewed data have a longer right tail.
- **Kurtosis / tails:** describes tail weight and concentration relative to a reference shape.

5.3 Important Discrete Distributions

Distribution	What it models	Parameters	Example
Bernoulli	One trial with two outcomes, 0 or 1	p = probability of success	One student passes or fails
Binomial	Number of successes in fixed n independent trials	n and p	Passes among 20 independent candidates

Distribution	What it models	Parameters	Example
Poisson	Number of events in a fixed exposure	lambda = expected count	Calls received per hour

FORMULAS

Bernoulli mean = p . Binomial mean = $n \cdot p$. Poisson mean = lambda and variance = lambda.

BINOMIAL CONDITIONS

Fixed number of trials, two outcomes, independent trials and constant success probability.

5.4 Important Continuous Distributions

Distribution	Shape / support	Main use	Parameters
Uniform	Flat over $[a,b]$	Values equally likely over a fixed range	Minimum a , maximum b
Normal	Symmetric bell shape	Measurements and sampling distributions under suitable conditions	Mean μ , standard deviation σ
Exponential	Right-skewed, $x \geq 0$	Waiting time between constant-rate events	Rate lambda; mean $1/\lambda$

FORMULA

z-score = $(x - \mu) / \sigma$. It states how many standard deviations a value lies above or below the mean.

For an approximately normal distribution, about 68% of observations lie within 1 standard deviation of the mean, about 95% within 2, and about 99.7% within 3. This rule is approximate and should not be used for strongly non-normal data.

5.5 How to Choose a Distribution

Data situation	Candidate distribution	Question to check
One yes/no outcome	Bernoulli	Are there exactly two states?
Successes in fixed trials	Binomial	Are trials independent with stable p ?
Event count per time/space	Poisson	Is a stable average rate plausible?
Symmetric measurement	Normal	Is the shape roughly bell-like without extreme tails?
Waiting time	Exponential	Is the event rate approximately constant?
Random value within fixed limits	Uniform	Are equal-length intervals equally likely?

Python: probability distributions with SciPy

```
from scipy import stats

# Probability of exactly 8 successes in 10 trials, p=0.8
print(stats.binom.pmf(8, n=10, p=0.8))

# Probability of at most 2 events when average count is 3
print(stats.poisson.cdf(2, mu=3))

# Probability that Normal(mean=60, sd=8) is below 70
print(stats.norm.cdf(70, loc=60, scale=8))
```

6. Parametric and Non-Parametric Methods

6.1 Parametric Methods

Parametric methods assume that data follow a distribution described by a small number of parameters, often a normal distribution with a mean and variance. When assumptions are reasonable, these methods can be efficient and powerful.

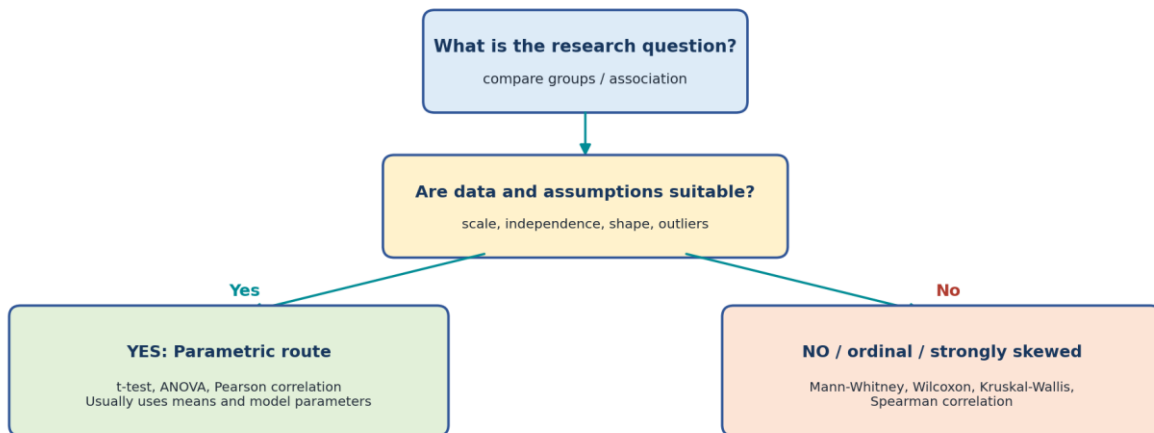
- Often use numeric values directly and focus on means or model coefficients.
- May assume normality, independence, linearity or equal variances.
- Examples: z-test, t-test, ANOVA, Pearson correlation and linear regression.

6.2 Non-Parametric Methods

Non-parametric methods make fewer or weaker distributional assumptions. Many use ranks, signs, medians or counts instead of raw numeric values. They are useful for ordinal data, strong skewness, extreme outliers or small samples when parametric assumptions are doubtful.

- Examples: Mann-Whitney U, Wilcoxon signed-rank, Kruskal-Wallis, Spearman rank correlation and chi-square test.
- They are not completely assumption-free; independence and suitable sampling still matter.
- They may answer a different question from their parametric counterpart.

CHOOSING BETWEEN PARAMETRIC AND NON-PARAMETRIC METHODS



Do not choose a test only from habit. Match the method to the question, measurement scale and assumptions.

Figure 4: A practical decision path for choosing a statistical method.

Point	Parametric	Non-parametric
Distribution assumptions	Usually stronger	Usually fewer or weaker
Data often used	Interval/ratio numeric data	Ordinal, ranks, signs or categories
Main summaries	Means, variances, coefficients	Medians, ranks, counts
Power when assumptions hold	Often higher	May be lower for the same parametric target
Robustness to skew/outliers	Can be sensitive	Often more robust
Examples	t-test, ANOVA, Pearson	Mann-Whitney, Kruskal-Wallis, Spearman

6.3 Common Method Pairs

Research situation	Parametric method	Non-parametric alternative
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Research situation	Parametric method	Non-parametric alternative
Compare two independent groups	Independent-samples t-test	Mann-Whitney U test
Compare paired measurements	Paired t-test	Wilcoxon signed-rank test
Compare 3 or more independent groups	One-way ANOVA	Kruskal-Wallis test
Measure association between two numeric variables	Pearson correlation	Spearman rank correlation

CAUTION

Do not select a test only by checking whether $p > 0.05$ in a normality test. Also examine plots, sample size, outliers, study design and the exact question.

Python: parametric and non-parametric comparison

```
from scipy import stats

group_a = [62, 65, 68, 70, 72]
group_b = [58, 60, 61, 63, 66]

print(stats.ttest_ind(group_a, group_b, equal_var=False))
print(stats.mannwhitneyu(group_a, group_b, alternative='two-sided'))
```

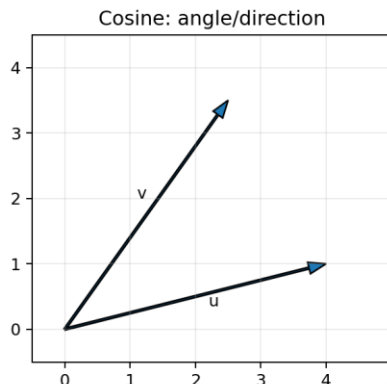
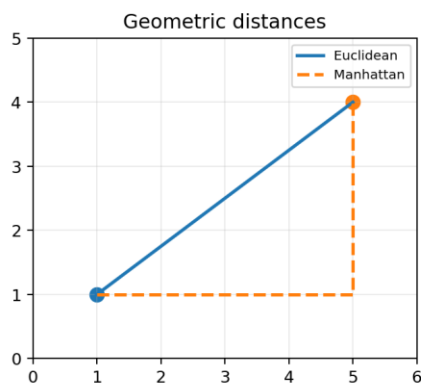
7. Distance Metrics

7.1 Meaning of Distance and Similarity

A distance metric measures how different two data points are. A small distance means the points are close according to the chosen definition. Distance is important in K-nearest neighbours, clustering, recommendation, anomaly detection and information retrieval.

KEY POINT

There is no universally best distance. The correct metric depends on feature type, scale and the meaning of similarity in the problem.

DISTANCE METRICS MEASURE DIFFERENT NOTIONS OF “CLOSENESS”**Binary/set distances****Hamming**

fraction or count of positions that differ

Jaccard

$1 - \text{intersection} / \text{union}$ for presence sets

Figure 5: Different distance measures can view the same pair of points differently.

7.2 Euclidean Distance

FORMULA	$d(x,y) = \sqrt{\sum((x_i - y_i)^2)}$
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For $A=(1,2)$ and $B=(4,6)$: $d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5$. Euclidean distance represents straight-line distance.

7.3 Manhattan Distance

FORMULA	$d(x,y) = \sum(x_i - y_i)$
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For $A=(1,2)$ and $B=(4,6)$: $d = |4-1| + |6-2| = 3+4 = 7$. It resembles moving along city blocks.

7.4 Minkowski and Chebyshev Distance

- **Minkowski distance:** a general family controlled by p . $p=1$ gives Manhattan and $p=2$ gives Euclidean.
- **Chebyshev distance:** the largest absolute coordinate difference. For A and B above, $\max(3,4)=4$.

7.5 Cosine Distance

Cosine similarity compares the angle between vectors and focuses on direction rather than magnitude. It is common for text vectors. Cosine distance is commonly defined as $1 - \text{cosine similarity}$.

FORMULA	$\text{cosine similarity} = (x \cdot y) / (\ x\ * \ y\)$; cosine distance = $1 - \text{similarity}$
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7.6 Hamming and Jaccard Distance

Metric	Best suited for	Meaning	Example
Hamming	Equal-length binary or categorical sequences	Fraction or count of positions that differ	10110 vs 10011 differ at 2 positions
Jaccard	Sets or binary presence/absence data	$1 - \text{intersection/union}$	$A=\{a,b,c\}$, $B=\{b,c,d\}$: similarity= $2/4=0.5$

7.7 Mahalanobis Distance

Mahalanobis distance accounts for feature scale and correlation. A movement along a naturally variable direction may count less than the same movement along a stable direction. It requires a reliable covariance estimate and is more advanced than ordinary Euclidean distance.

Metric	Core idea	Typical use
Euclidean	Straight-line numeric distance	KNN and clustering on scaled numeric data
Manhattan	Sum of coordinate differences	High-dimensional or grid-like movement
Cosine	Difference in direction	Text and sparse vectors
Hamming	Mismatched positions	Binary codes or sequences
Jaccard	Difference in set membership	Binary presence/absence or item sets
Mahalanobis	Scale- and correlation-aware distance	Multivariate anomaly detection

EXAM TIP

Scale numeric features before using Euclidean or Manhattan distance; otherwise a feature measured in large units can dominate the result.

Python: common distance measures

```
import numpy as np
from scipy.spatial import distance

x = np.array([1, 2])
y = np.array([4, 6])
print('Euclidean:', distance.euclidean(x, y))
print('Cityblock:', distance.cityblock(x, y))
print('Cosine distance:', distance.cosine(x, y))

A = {1, 2, 3}
B = {2, 3, 4}
jaccard_similarity = len(A & B) / len(A | B)
print('Jaccard distance:', 1 - jaccard_similarity)
```

8. Exploratory Data Analysis: Basic Tools

8.1 What is EDA?

Exploratory Data Analysis (EDA) is the systematic process of examining a dataset before formal modelling. It uses summaries, plots and questions to understand data quality, distributions, relationships, unusual values and possible explanations.

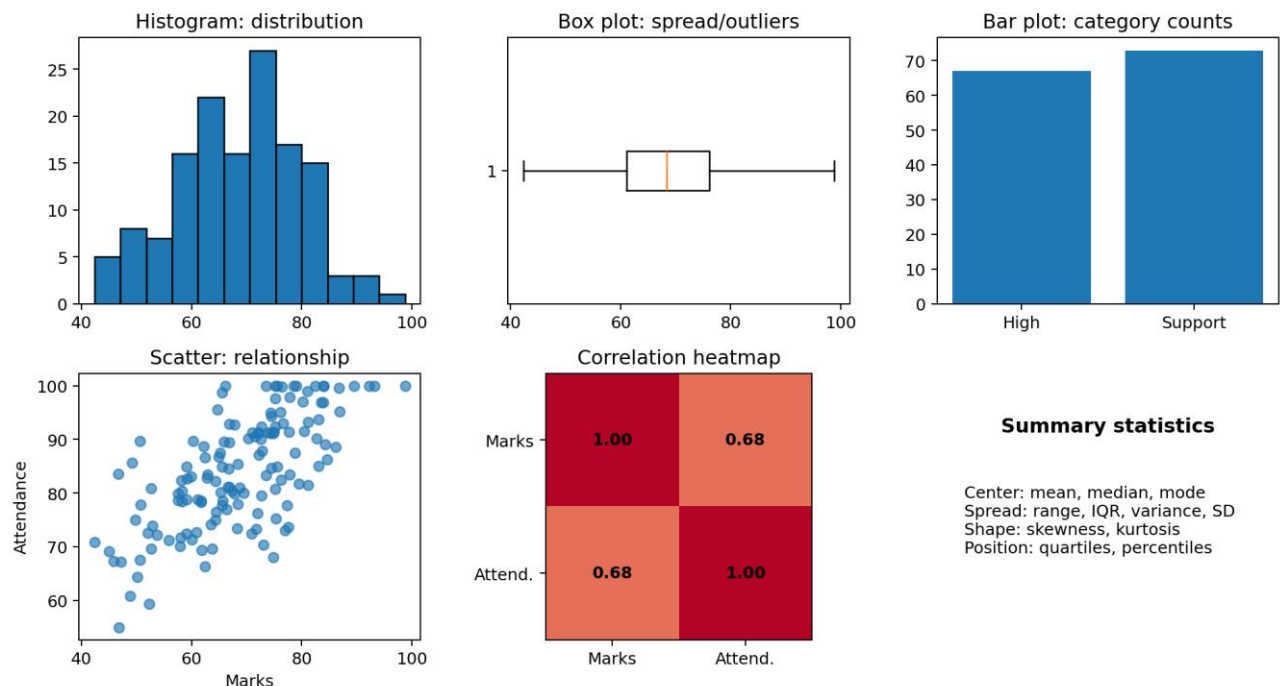
BASIC EDA TOOLBOX: NUMBERS AND VISUALS WORK TOGETHER

Figure 6: EDA combines data-quality checks, numerical summaries and multiple visual views.

8.2 Questions Asked During EDA

- How many rows and columns are present?
- What does each variable mean and what is its data type?

- Are values missing, duplicated, impossible or inconsistent?
- What are the centre, spread, shape and unusual values of each variable?
- How do variables relate to one another?
- Do patterns change across groups or time?
- Could a pattern be caused by data collection, leakage or a third variable?

8.3 Summary Statistics

Measure	Meaning	Sensitive to outliers?
Count	Number of non-missing observations	No
Mean	Arithmetic average	Yes
Median	Middle ordered value	Less sensitive
Mode	Most frequent value/category	Usually less sensitive
Range	Maximum - minimum	Very sensitive
Variance / standard deviation	Average squared spread / its square root	Sensitive
Interquartile range (IQR)	Q3 - Q1; spread of middle 50%	More robust
Percentile	Value below which a stated percentage lies	Depends on percentile

FORMULA

Mean = $\sum(x)/n$. Range = max-min. IQR = Q3-Q1. Sample variance = $\sum((x-x_{\text{bar}})^2)/(n-1)$.

KEY POINT

Use mean and standard deviation for roughly symmetric data without severe outliers. Use median and IQR for strongly skewed data or data with extreme values.

8.4 Univariate Plots

Plot	Variable type	What it helps reveal
Bar chart	Categorical	Counts or percentages by category
Histogram	Numeric	Distribution shape, centre, spread and modes
Box plot	Numeric	Median, IQR and potential outliers
Density plot	Continuous numeric	Smoothed distribution shape
Line plot	Time-ordered numeric	Trend, cycles and sudden changes

8.5 Bivariate and Multivariate Plots

Plot	Variables	Use
Scatter plot	Two numeric variables	Direction, strength, non-linearity and clusters
Grouped box plot	Numeric plus categorical	Compare distributions across groups
Stacked/grouped bar	Two categorical variables	Compare category proportions
Correlation heatmap	Several numeric variables	Scan pairwise linear correlations

Plot	Variables	Use
Pair plot	Several numeric variables	View many distributions and relationships
Faceted plot	Variables split by a group	Check whether a pattern changes by subgroup

CAUTION

Correlation does not prove causation. A relationship can arise from confounding, selection bias, reverse direction or coincidence.

8.6 A Basic EDA Sequence

1. Understand the meaning, unit and source of each variable.
2. Inspect shape, data types and a few rows.
3. Check missing values, duplicates and impossible values.
4. Summarize numeric and categorical variables.
5. Plot individual variable distributions.
6. Explore relationships relevant to the question.
7. Investigate unusual values and subgroup differences.
8. Record findings, limitations and next questions.

Python: a beginner EDA checklist

```
import pandas as pd
import matplotlib.pyplot as plt

# Replace with your file path
df = pd.read_csv('students.csv')

print(df.shape)
print(df.head())
print(df.info())
print(df.isna().sum())
print(df.describe(include='all'))

# Univariate plot
df['marks'].hist(bins=10)
plt.xlabel('Marks')
plt.ylabel('Frequency')
plt.show()

# Bivariate plot
df.plot.scatter(x='attendance', y='marks')
plt.show()
```

9. Philosophy of Exploratory Data Analysis

9.1 EDA is a Way of Thinking

EDA is not a fixed collection of charts. Its philosophy is to remain curious, use multiple views, question assumptions, search for surprises and let the data suggest better questions. The analyst should not force the data to confirm the first expectation.

- **Be curious:** ask what is unusual and why it may have occurred.
- **Use multiple representations:** a table, histogram and box plot can reveal different aspects of the same variable.
- **Be sceptical:** check whether a pattern could be caused by errors, missing data or selection.
- **Explore subgroups:** an overall trend can hide opposite trends inside groups.
- **Document decisions:** record filtering, transformations and exclusions so the work is reproducible.

- **Separate discovery from proof:** a pattern discovered during EDA should be validated with fresh data or a carefully designed confirmatory analysis.

9.2 Exploratory versus Confirmatory Analysis

Point	Exploratory analysis	Confirmatory analysis
Purpose	Discover patterns and generate hypotheses	Test a pre-specified hypothesis
Questions	Flexible and evolving	Defined before examining outcomes
Methods	Many summaries and visual views	Formal model/test with planned assumptions
Risk	Finding patterns by chance	Invalid inference if plan is changed after seeing results
Good practice	Label findings as exploratory	Use independent validation or pre-specified analysis

MEMORY AID

EDA asks: “What might be happening?” Confirmatory analysis asks: “Does planned evidence support this exact claim?”

9.3 Common EDA Mistakes

- Looking only at averages and ignoring distributions.
- Deleting outliers without investigating their cause.
- Using too many charts without a question.
- Ignoring missingness patterns and data collection processes.
- Treating correlation as proof of cause.
- Reporting an exploratory discovery as if it had been predicted in advance.
- Choosing axes, bins or colour scales that exaggerate a pattern.

10. The Data Science Process

10.1 Why a Process is Needed

A data science project is more than training an algorithm. It begins with a decision or problem, and it continues through data collection, analysis, modelling, communication, deployment and monitoring. The process is iterative: findings often require earlier steps to be revised.

THE DATA SCIENCE PROCESS IS ITERATIVE, NOT A ONE-WAY PIPELINE

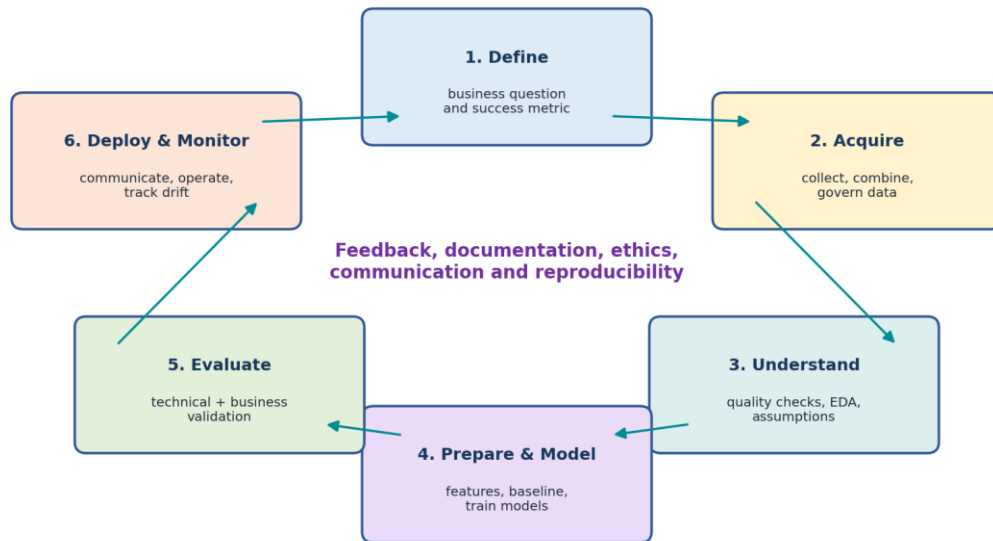


Figure 7: The data science process is iterative rather than a one-way pipeline.

10.2 Main Stages

Stage	Main questions	Typical output
1. Problem understanding	What decision is needed? Who will use the result? What is success?	Problem statement, scope, metric and constraints
2. Data collection	What data are needed? From where? With what permissions?	Source inventory and raw data
3. Data understanding	What do variables mean? How were they measured?	Data dictionary and quality report
4. Preparation	How should missing values, errors and formats be handled?	Clean analysis-ready dataset
5. EDA	What patterns, relationships and anomalies exist?	Findings, plots and new questions
6. Feature engineering	What useful representations can be created?	Model input features
7. Modelling	Which baseline and candidate models fit the objective?	Trained models
8. Evaluation	Does performance generalize and satisfy practical needs?	Metrics, error analysis and validation
9. Communication / deployment	How will findings or predictions be used?	Report, dashboard, application or API
10. Monitoring	Does data or performance change over time?	Alerts, retraining and maintenance plan

10.3 Problem Statement Template

TEMPLATE	For [user/stakeholder], use [available data] to [estimate/predict/understand] [outcome] so that [decision/action] improves, while respecting [constraints, fairness, privacy and cost].
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10.4 Baseline, Validation and Communication

- **Baseline:** a simple rule or model used as the minimum comparison. A complex model must beat a meaningful baseline.
- **Validation:** evaluate on data not used to fit the model and analyse errors across important groups.
- **Communication:** state the question, data, method, result, uncertainty, limitations and recommended action.
- **Monitoring:** watch for changing inputs, reduced performance, unfair impacts and operational failures.

KEY POINT

A technically accurate model can still fail if it answers the wrong question, uses biased data or cannot support a real decision.

10.5 Mini Case: Student Support

Process step	Example decision
Problem	Identify students who may need academic support early enough for useful intervention
Data	Attendance, assessment history, assignment submission and course context
EDA	Check missingness, class imbalance, subgroup patterns and leakage from future information
Model	Begin with a transparent baseline and a simple classification model
Evaluation	Use recall, precision and subgroup error analysis, not accuracy alone
Action	Offer supportive counselling or resources; never treat a prediction as proof
Monitoring	Check whether data and model performance change each semester

RESPONSIBLE USE

Predictions about people require privacy, fairness, human review and a beneficial intervention. A risk score should support—not replace—responsible human judgement.

11. Integrated Python Practical

11.1 Objective

Use a small student dataset to calculate summaries, create EDA plots, measure relationships, compare groups and compute distances. The example is for learning; conclusions from a real study require a sound sampling design and domain understanding.

Complete beginner practical

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
from scipy.spatial import distance

# 1. Create a small dataset
df = pd.DataFrame({
    'hours':    [2, 3, 4, 5, 6, 7, 8, 3, 6, 7],
    'attendance': [58, 64, 70, 76, 82, 88, 92, 68, 84, 90],
    'marks':    [42, 48, 55, 63, 69, 78, 86, 51, 72, 81],
    'group':    ['A','A','A','A','A','B','B','B','B','B']
})

# 2. Basic understanding and summaries
print(df.head())
print(df.info())
print(df.describe())
print(df.groupby('group')['marks'].agg(['count','mean','median','std']))

# 3. Univariate EDA
df['marks'].hist(bins=5, edgecolor='black')
plt.title('Distribution of Marks')
plt.xlabel('Marks'); plt.ylabel('Frequency')
plt.show()

# 4. Bivariate EDA
df.plot.scatter(x='attendance', y='marks')
plt.title('Attendance and Marks')
plt.show()

# 5. Correlation (linear association)
print('Pearson:', stats.pearsonr(df['attendance'], df['marks']))
print('Spearman:', stats.spearmanr(df['attendance'], df['marks']))

# 6. Compare two independent groups
A = df.loc[df['group']=='A', 'marks']
B = df.loc[df['group']=='B', 'marks']
print("Welch t-test:", stats.ttest_ind(A, B, equal_var=False))
print('Mann-Whitney:', stats.mannwhitneyu(A, B, alternative='two-sided'))

# 7. Distance between first two students using hours and attendance
x = df.loc[0, ['hours','attendance']].to_numpy(dtype=float)
y = df.loc[1, ['hours','attendance']].to_numpy(dtype=float)
print('Euclidean:', distance.euclidean(x, y))
print('Manhattan:', distance.cityblock(x, y))
```

11.2 How to Interpret the Output

1. Use describe() to check centre, spread and possible range problems.
2. Use the histogram to see whether marks are symmetric, skewed, clustered or unusual.
3. Use the scatter plot to inspect direction and form before interpreting correlation.
4. Pearson correlation measures linear association; Spearman measures monotonic rank association.
5. A group test result must be interpreted together with group summaries, effect size, assumptions and study design.
6. The distance calculation is dominated by attendance because attendance has a larger numeric scale; scaling would be needed for fair comparison.

PRACTICE

Modify one extreme mark, rerun summaries and compare how the mean, median, Pearson correlation and Spearman correlation change.

12. Quick Revision, Formula Sheet and Glossary

12.1 One-Line Revision

Concept	One-line meaning
Population	Complete group about which a conclusion is required
Sample	Observed subset used as evidence
Parameter	Population numerical value
Statistic	Sample numerical value
Inference	Learning about a population from sample data
Probability distribution	Possible random-variable values and their probabilities
Parametric method	Method based on a specified distributional/model form
Non-parametric method	Method using fewer distribution assumptions, often ranks or counts
Distance metric	Rule for quantifying difference between points
EDA	Question-driven exploration using summaries and visualizations
Data science process	Iterative path from problem definition to monitoring

12.2 Formula Sheet

Formula	Meaning
Mean = $\sum(x) / n$	Arithmetic average
Range = maximum - minimum	Total observed spread
IQR = $Q3 - Q1$	Spread of the middle 50%
Sample variance = $\sum((x - \bar{x})^2) / (n - 1)$	Average squared sample spread
SE(mean) = $s / \sqrt{n}$	Estimated uncertainty of sample mean
Confidence interval = estimate $\pm$ critical value $\times$ SE	Estimate with margin of error
$z = (x - \mu) / \sigma$	Standardized position
Euclidean = $\sqrt{\sum((x_i - y_i)^2)}$	Straight-line distance
Manhattan = $\sum(x_i - y_i)$	City-block distance
Cosine distance = $1 - \text{cosine similarity}$	Difference in vector direction
Jaccard similarity = intersection/union	Overlap between sets

12.3 Glossary

Term	Student-friendly meaning
Assumption	A condition required for a method or interpretation
Bias	Systematic error that pushes results in a direction
Confidence interval	A sample-based range showing estimate precision

Term	Student-friendly meaning
Confounder	A third variable related to both an apparent cause and outcome
Distribution	Pattern of possible values and their frequency/probability
Effect size	Magnitude of a difference or relationship
Hypothesis	A testable statement about a population or process
Outlier	An observation unusually far from most others
p-value	Compatibility measure between data and H_0 under test assumptions
Robust method	A method less affected by departures or extreme values
Skewness	Asymmetry in a distribution
Statistical power	Probability of detecting a real effect under stated conditions

13. Self-Assessment

13.1 Multiple-Choice Questions

1. A number calculated from a sample is called a:

- A. Parameter
- B. Statistic
- C. Census
- D. Population

2. Which sampling method ensures planned representation from important groups?

- A. Convenience
- B. Volunteer
- C. Stratified
- D. Cluster only

3. Which measure is usually most robust to an extreme outlier?

- A. Mean
- B. Median
- C. Range
- D. Variance

4. A binomial distribution models:

- A. Waiting time
- B. Successes in fixed binary trials
- C. Continuous temperature
- D. Any category

5. A small p-value is best described as:

- A. The probability H_0 is false
- B. Stronger incompatibility with H_0 under assumptions
- C. Proof of practical importance
- D. The effect size

6. Which test is a non-parametric alternative to an independent t-test?

- A. Pearson correlation
- B. Mann-Whitney U
- C. Paired t-test
- D. Linear regression

7. Which distance focuses primarily on vector direction?

- A. Euclidean
- B. Manhattan
- C. Cosine
- D. Hamming

8. A histogram is mainly used for:

- A. One numeric variable
- B. Two categorical variables only
- C. Text paragraphs
- D. Set intersection

9. EDA findings discovered after many searches should usually be:

- A. Treated as final proof
- B. Hidden
- C. Validated on fresh evidence
- D. Converted directly to causation

10. The first stage of a data science process should be:

- A. Choose deep learning
- B. Define the problem and decision
- C. Deploy a dashboard
- D. Remove all outliers

ANSWER KEY

1-B, 2-C, 3-B, 4-B, 5-B, 6-B, 7-C, 8-A, 9-C, 10-B

13.2 Very Short and Short Questions

1. Define population and sample with one example.
2. Differentiate a parameter and a statistic.
3. What is the difference between sampling error and bias?
4. State four conditions of a binomial distribution.
5. What does a standard error measure?
6. What is a p-value? Mention one common misinterpretation.
7. Differentiate parametric and non-parametric methods.
8. Calculate Euclidean and Manhattan distance between (2,3) and (5,7).
9. When is cosine distance more useful than Euclidean distance?
10. Differentiate a histogram and a bar chart.
11. State five questions asked during EDA.
12. Why is the data science process iterative?

13.3 Long-Answer Questions

1. Explain population, sample, sampling frame, parameter, statistic, sampling methods, sampling error and bias with examples.
2. Explain point estimation, standard error, confidence intervals, hypothesis testing, p-values and Type I/Type II errors.
3. Discuss the main discrete and continuous probability distributions and give a suitable application of each.
4. Compare parametric and non-parametric methods and explain how you would choose between them.
5. Explain Euclidean, Manhattan, Minkowski, cosine, Hamming, Jaccard and Mahalanobis distances.
6. Describe the tools and philosophy of EDA. Explain how exploratory analysis differs from confirmatory analysis.
7. Explain every stage of the data science process using one real-world case study.

13.4 Practical Exercises

1. Create a pandas DataFrame with 20 student records. Produce shape, data types, missing-value counts and descriptive summaries.
2. Plot a histogram and box plot of marks. Write three observations about centre, spread and unusual values.
3. Create a scatter plot of study hours and marks. Calculate Pearson and Spearman correlation and interpret them carefully.
4. Compare marks of two classes using summaries, box plots, a t-test and Mann-Whitney test. State assumptions and limitations.
5. Calculate Euclidean, Manhattan and cosine distances for three student feature vectors before and after scaling.
6. Choose a public dataset and prepare a one-page EDA report containing the problem, data quality, summaries, plots, findings and limitations.